

Solution Requirements

Date	17 JULY 2026
Team ID	SWTID-2026-4144
Project Name	RISING WATERS
Maximum Marks	4 Marks

Step 1: Functional Requirements

S.No	Category	Requirement Description	Priority
1	Authentication	Users, disaster management officials and administrators must securely log in via OTP.	High
2	Authorization	Provide role-based access for Admin, Field Officer and Public User with different permissions.	High
3	External Interfaces	Integrate weather forecast APIs, river-level sensors, GPS services and SMS gateways.	High
4	Transaction Processing	Collect, validate and store flood sensor data, user reports and emergency notifications in real-time.	High
5	Reporting	Generate flood-risk reports, water-level history, affected-area summaries and interactive dashboards.	Medium
6	Business Rules	Automatically issue alerts when water level crosses predefined thresholds at critical locations.	High
7	Compliance	Follow disaster-management standards, data privacy policies and secure data handling practices.	Medium
8	Other	Allow multilingual support, offline data capture and notification history.	Low

Step 2: Non-Functional Requirements

S.No	Category	Requirement Description	Acceptance Criteria
1	Performance	Flood alerts should be processed quickly.	
2	Scalability	System should support increasing users and sensor devices.	10,000+ concurrent users
3	Security	Encrypt sensitive information and secure communication.	AES-256 encryption & HTTPS
4	Reliability	Service should remain available during emergencies.	99.9% uptime
5	Usability	Simple, mobile-friendly interface with accessible design.	Easy navigation for all users
6	Maintainability	Code should be modular and easy to update.	Quick deployment of updates